

Geometric Energy Landscapes for Dual-Target CAR-T Marker Discovery in Endometriosis

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Abstract

Endometriosis is common and debilitating. When symptoms are severe or refractory, surgery is warranted, yet clinical outcomes are limited by poor lesion visibility, incomplete resection, and high recurrence. These constraints motivate a more specific therapeutic approach. Dual-target CAR-T cells can require co-expression of two surface markers, offering greater selectivity than single-marker targeting, and CAR-T targeting of activated fibroblasts in fibrotic disease provides a relevant precedent. Here we present a new differentiable programming algorithm that uses single-cell transcriptomics to identify optimal marker pairs by principled search rather than ad hoc choice. We cast pair selection as a polynomial-time energy minimization problem on a non-convex geometric surface, obtained by relaxing the combinatorial AND objective into a smooth log-space energy. In practice the optimization converges rapidly and consistently across seeds, suggesting shallow local minima and an identifiable optimum. The approach efficiently discovers lesion-enriched marker pairs that are biologically plausible and robust across patients, providing a practical path toward dual-target CAR designs in an underfunded disease area.

Introduction

Endometriosis is characterized by ectopic endometrial-like tissue that is difficult to delineate surgically and prone to recurrence. Current therapies often trade efficacy for systemic side effects, and surgical resection remains the default once hormonal management fails. A dual-target CAR-T approach offers a compelling path: requiring co-expression of two lesion-specific surface markers could reduce off-target toxicity while preserving potency, and dual-target CAR-T has clinical precedent in B-cell lymphoma [1].

The computational challenge is to search a combinatorial space of marker pairs while respecting biological constraints: marker co-expression should be high in lesions, low in healthy uterine and off-target tissues, and robust across patients [2]. Concretely, with G candidate surface genes, the number of unordered pairs is $\binom{G}{2} = G(G-1)/2$. Even with a conservative list of $\sim 3,000$

surface proteins, this yields about 4.5 million pairs, before patient stratification, tissue filters, and robustness checks. Incorporating these constraints and stability analyses makes brute-force screening impractical at scale. We therefore present a differentiable programming algorithm that relaxes the NP-hard grid search into a polynomial-time energy minimization over a smooth, non-convex landscape, framing marker discovery as geometry on a data-driven energy surface rather than a brute-force enumeration.

Methods

Input and binarization

Let $X \in \{0, 1\}^{N \times G}$ be a binarized single-cell expression matrix for N cells and G genes, with labels $y_i \in \{0, 1\}$ indicating lesion (1) or healthy (0). To reduce confounding from immune infiltrates, cells are annotated with cell type labels (using CellTypist and the Human Endometrium Atlas) and immune lineages are filtered prior to analysis [3, 4]. We denoise with scVI and include sample as the batch covariate to reduce technical effects [5]. Binarization is performed after denoising and thresholding at a fixed normalized count per 10k reads. We defined gene expression as > 1 normalized count per 10k library, a commonly used threshold approximating consistent transcript detection in scRNA-seq data. To control for patient sampling bias, specificity scores are computed with patient-level stratification. Immune cells are filtered rather than pooled with healthy stromal cells to avoid dominating the signal, and robustness is assessed over multiple random seeds.

Learnable soft selection

For each gene j , a learnable logit α_j encodes selection. We sample a Gumbel-Softmax mask

$$m_j = \sigma \left(\frac{\alpha_j + g_j}{\tau} \right), \quad m_j \in (0, 1),$$

where g_j is Gumbel noise and τ is a temperature. This Gumbel-Softmax relaxation provides low-variance gradients for discrete selection and enables efficient optimization under the two-marker constraint [6, 5]. At low τ , m_j becomes nearly binary.

Differentiable AND

To model a dual-marker AND, we define per-cell activation as

$$a_i = \prod_j (1 - m_j(1 - X_{ij})).$$

If a selected marker is absent ($X_{ij} = 0$ and $m_j \approx 1$), the product collapses to 0; if all selected markers are present, $a_i \approx 1$. For numerical stability, we optimize in log-space:

$$\log a_i = \sum_j \log (1 - m_j(1 - X_{ij})).$$

Because X_{ij} is binary, this simplifies to

$$\log a_i = \sum_j (1 - X_{ij}) \log(1 - m_j),$$

a sum of penalties over missing selected markers.

Objective with cardinality constraint

We encourage high activation in lesions and low activation in healthy cells:

$$\mathcal{L}_{\text{act}} = -E_{\text{lesion}}[\log a_i + \epsilon] + E_{\text{healthy}}[a_i].$$

To enforce exactly two markers, we add a cardinality penalty:

$$\mathcal{L}_{\text{card}} = \lambda \left(\sum_j m_j - 2 \right)^2,$$

and optimize $\mathcal{L} = \mathcal{L}_{\text{act}} + \mathcal{L}_{\text{card}}$ by gradient descent. At inference, we take the top two α_j as the final pair.

Results

Initial experiments on the endometriosis atlas (patient-stratified, immune cells filtered via CellTypist with Human Endometrium Atlas labels) identify a stable two-gene pair with strong lesion enrichment across multiple random seeds [2]. In a held-out test split, the pair GAS1 + BST2 shows approximately 10 $\times$ enrichment for lesion activation while remaining suppressed in healthy samples. Separately, FAP emerges as a top-ranked marker, consistent with prior targeting of activated fibroblasts in fibrotic disease contexts [7].

These results suggest that differentiable AND-based subset selection can recover biologically plausible, lesion-specific marker combinations without exhaustive brute force. The next step is to validate candidate pairs against broader tissue atlases (human and murine) to quantify off-target risk, and to assess feasibility of available scFvs for engineering dual CAR constructs [8].

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